

§2-8 Derivatives

Homework : 4,7,15,17,19,29,35,36

Definition : f has a derivative at $x = a$ (or f is differentiable at $x = a$) provided

that $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$ exists or equivalently $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$.

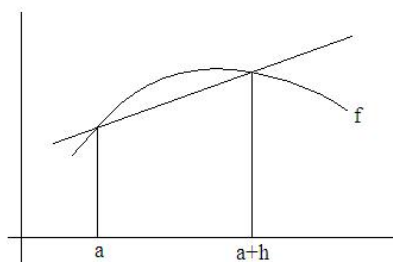
Its limit will be denoted by $f'(a)$.

Definition : The derivative of $y = f(x)$ is defined to be

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad \text{whenever the limit exists.}$$

Remark : Geometrically,

(i) $\frac{f(a+b) - f(a)}{b}$ = the slope of the secant.



(ii) $f'(a)$ = the slope of the tangent of $y = f(x)$ at $x = a$.

Generally speaking : $f'(a)$ = the rate of change of f at a .

(i) 若 $f(t)$ = positron at time t .

$$\Rightarrow f'(a) = \text{instantaneous velocity at time } t = a.$$

(ii) $f(x)$ = cost to produce x amount of fabric.

$$\Rightarrow f'(a) = \text{marginal cost when the product level } x = a.$$

Example 1 : $f(x) = x^2 - 8x + 9$. Compute $f'(a)$ by definition.

Solution :

$$\begin{aligned} & \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \\ &= \lim_{x \rightarrow a} \frac{x^2 - 8x + 9 - a^2 + 8a - 9}{x - a} \\ &= \lim_{x \rightarrow a} (x + a - 8) \\ &= 2a - 8 \end{aligned}$$

Example 2 : Find an equation of the tangent to $y = x^2 - 8x + 9$ at $(3, -6)$.

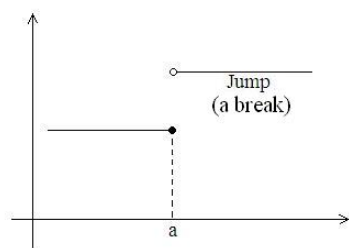
Solution : Slope $f'(3) = -2$

Point $(3, -6)$

$$\Rightarrow y + 6 = -2(x - 3)$$

When does the derivative of f at $x = 2$ not exist?

(i)

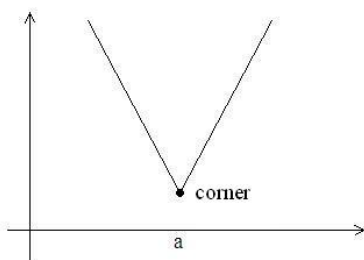


f has break at $x = a$.

例如： $f(x) = [x]$.

f has no derivatives at all the integers.

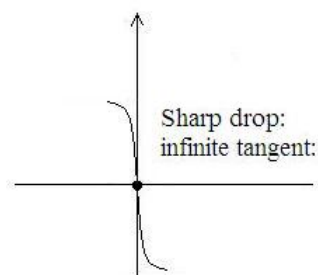
(ii)



例： $f(x) = |x|$

f has continuous at $x = 0$,
but is not differentiable at $x = 0$.

(iii)



例： $f(x) = x^{\frac{1}{3}}$

- Roughly speaking, if f has a break, or a corner or a sharp turn at $x = a$, then f has no derivative there.

Theorem : f has derivative at $x = a$
 $\Rightarrow f$ is continuous at $x = a$.

Example 3 : $\lim_{x \rightarrow 0} \frac{3^x - 1}{x} = f'(0)$. Find f .

Solution : $f(x) = 3^x$.

Example 4 : $f(x) = \begin{cases} x \sin \frac{1}{x}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$

Discuss the differentiability and continuity of f at $x = 0$.

Solution :

(i) f is continuous at $x = 0$ by Sandwich Theorem.

(ii) $\lim_{x \rightarrow 0} \frac{f(x) - 0}{x - 0} = \lim_{x \rightarrow 0} \sin \frac{1}{x}$ 不存在.

$\Rightarrow f$ is not differentiable at $x = 0$ either.

Example 5 : $f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$

Solution :

$$f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - 0}{x - 0} = \lim_{x \rightarrow 0} \sin \frac{1}{x} = 0$$

$\Rightarrow f$ is differentiable at $x=0$ (and hence continuous at $x=0$.)

Example 6 : $c = f(v)$: v : speed $\left(\frac{\text{miles}}{\text{hr}}\right)$

c : gas consumption $\left(\frac{\text{gallons}}{\text{hr}}\right)$

(i) What is the meaning of the derivative $f'(v)$?

What are its units?

(ii) Write a sentence that explains the meaning of the equation

$$f'(20) = -0.05.$$

Solution :

(i) $f'(v)$ = the consumption rate of a car at the speed v .

$$\text{Its unit} = \frac{\frac{\text{gallons}}{\text{hr}}}{\frac{\text{miles}}{\text{hr}}} = \frac{\text{gallons}}{\text{mile}}.$$

(ii) The fuel consumption is decreasing by $0.05 \frac{\text{gallons}}{\text{mile}}$ as the

car's reaches $20 \frac{\text{miles}}{\text{hr}}$. That is, if you increase your speed at

$21 \frac{\text{miles}}{\text{hr}}$, you could expect to decrease your fuel consumption

by about $0.05 \frac{\text{gallons}}{\text{mile}}$.

Example 7 : Let $f(x) = \sqrt{ax+b}$, $f(0)=1$ and $f'(0)=1$. Find a and b

Solution :

$$f(0) = 1 = \sqrt{b} \Rightarrow b = 1$$

$$\begin{aligned}
1 &= \lim_{x \rightarrow 0} \frac{\sqrt{ax+1}-1}{x} \\
&= \lim_{x \rightarrow 0} \frac{ax(\sqrt{ax+1}+1)}{x} \\
&= 2a \\
\Rightarrow a &= \frac{1}{2}
\end{aligned}$$